Comparison of AlexNet and DHVT on the Same Image Classification Dataset

Course Presentation

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Outline

1. Introduction
2. Dataset and Workflow
3. Models
4. Training and Difficulties
5. Results
6. Conclusion

Project Goal

- The project compares a classical convolutional model and a transformer-style model under the same dataset setting.
- The two models are trained, saved, reloaded, and evaluated in the same notebook workflow.
- The main objective is to analyze not only accuracy, but also training behavior, confidence, and difficult examples.

Why Compare AlexNet and DHVT?

Model	Role in the study
AlexNet	CNN baseline with strong image inductive bias and a clear layer-by-layer structure.
DHVT	Transformer-style model adapted to small datasets through overlapped patch embedding and DAFF blocks.

- This comparison contrasts local convolutional processing with token-based attention.
- Both models use the same dataset configuration to keep the experiment fair.

Dataset Options

Option	Description
`CIFAR-10`	Standard 10-class benchmark used as the main baseline dataset.
`CIFAR-100`	Same image format with more classes and higher difficulty.
`custom_folder`	User dataset with `train/class_name` and `val/class_name` folders.

- Both models can be trained on the same selected dataset.
- Custom inference images are stored separately in `test_images/`.

Shared Experimental Workflow

Step	Procedure
1	Choose local runtime or Colab runtime.
2	Select the dataset and preprocessing settings.
3	Train AlexNet and DHVT on the same split.
4	Save the best checkpoints for both models.
5	Reload the saved models for evaluation and inference.
6	Compare training curves and qualitative outputs.

AlexNet

- AlexNet uses stacked convolutions, ReLU, pooling, and fully connected layers.
- In the notebook, it serves as the CNN baseline for the comparison.
- The convolution operation is written as:

$$Y_{i,j} = \sum_u \sum_v K_{u,v} X_{i+u,j+v}$$

$$\mathrm{ReLU}(x) = \max(0, x)$$

- This structure emphasizes local feature extraction and spatial hierarchy.

DHVT

- DHVT is a transformer-style architecture adapted to smaller datasets.
- The model includes SOPE, class token, positional embeddings, attention blocks, and DAFF.
- Patch-token embedding can be summarized as:

$$X \in \mathbb{R}^{B \times C \times H \times W} \rightarrow E \in \mathbb{R}^{B \times N \times d}$$

$$Z_0 = [x_{\text{cls}}; E], \qquad Z = Z_0 + P$$

- The model is trained from scratch rather than used only for fine-tuning.

Attention and DHVT Block

$$Q = ZW_Q, \qquad K = ZW_K, \qquad V = ZW_V$$

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

$$Z' = Z + \mathrm{MHSA}(\mathrm{LN}(Z))$$

$$Z'' = Z' + \mathrm{DAFF}(\mathrm{LN}(Z'))$$

- The DAFF block adds a convolution-aware feed-forward stage.
- This design helps the transformer preserve useful locality for small datasets.

Training Strategy

- Both models are trained under the same dataset configuration.
- Checkpoints are saved and reloaded inside the notebook.
- The objective function is the cross-entropy loss:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log(\hat{p}_c)$$

- Validation accuracy is monitored to keep the best weights for each model.

Main Difficulties and Solutions

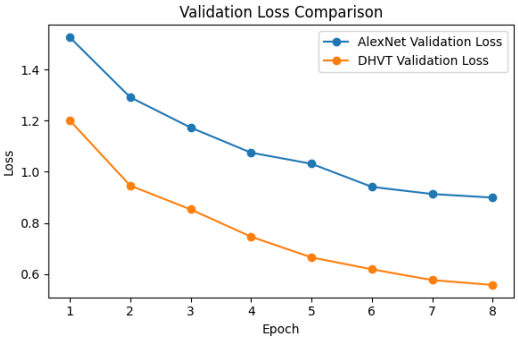
Difficulty	Solution
Fair comparison	Use one notebook and one shared dataset configuration.
Small-dataset transformer training	Use a DHVT-style architecture trained from scratch.
Homework-style presentation	Keep the workflow step-by-step in the notebook.
Reusability	Support multiple datasets and custom folders.
Long training runs	Print intermediate progress during training.

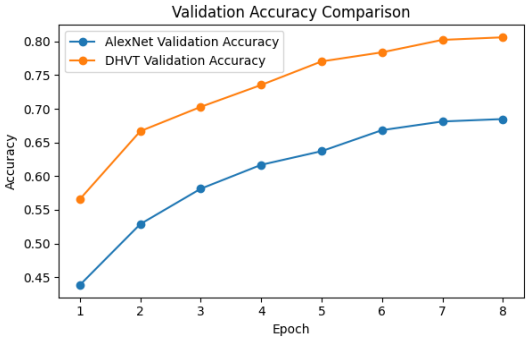
Main Numerical Results

Metric	AlexNet	DHVT
Best validation accuracy	0.6848	0.8062
Number of parameters	58,322,314	6,504,202
Macro precision	0.6811	0.8057
Macro recall	0.6848	0.8062
Macro F1-score	0.6814	0.8056

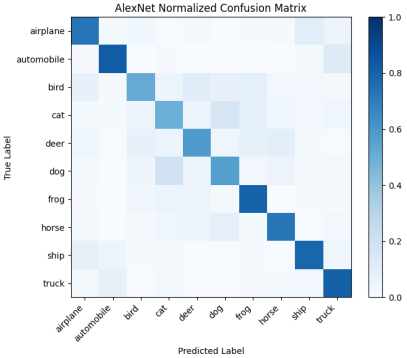
- DHVT is both more accurate and more parameter-efficient.
- The advantage is confirmed by several metrics, not by accuracy alone.

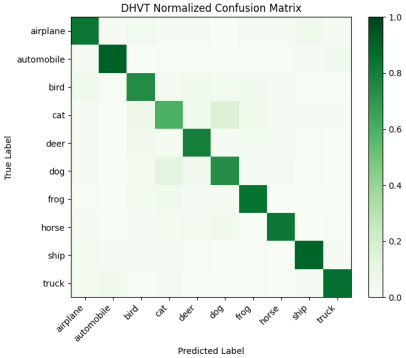
Validation Curves





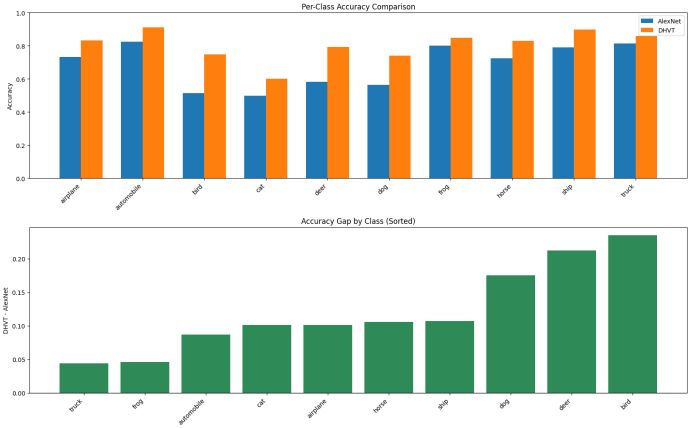
Confusion Matrices





- DHVT has a stronger diagonal, which means better class-level consistency.
- Both models still confuse visually similar classes such as cat and dog.

Per-Class Accuracy



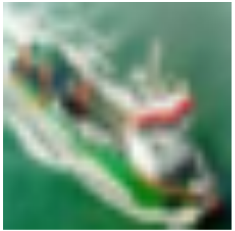
- The gain of DHVT is broad across classes rather than concentrated in one category.
- The biggest improvements appear on visually difficult classes such as bird, deer, and dog.

Misclassified Examples

True: ship
AlexNet: automobile
DHVT: ship

True: cat
AlexNet: dog
DHVT: cat

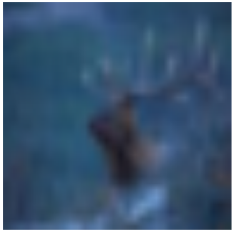
True: ship
AlexNet: frog
DHVT: frog



True: horse
AlexNet: truck
DHVT: horse

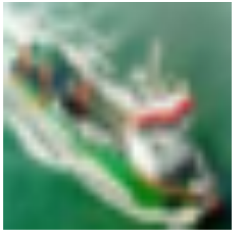
True: horse
AlexNet: horse
DHVT: dog

True: deer
AlexNet: airplane
DHVT: bird

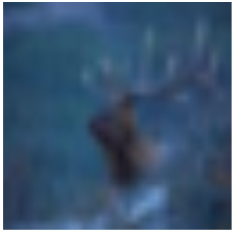


Both Models Wrong

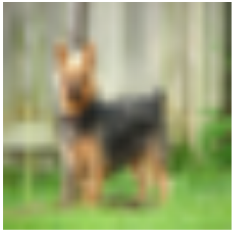
True: ship
AlexNet: frog
DHVT: frog



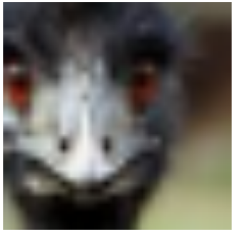
True: deer
AlexNet: airplane
DHVT: bird



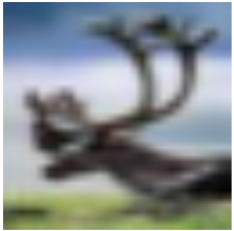
True: dog
AlexNet: deer
DHVT: deer



True: bird
AlexNet: cat
DHVT: automobile

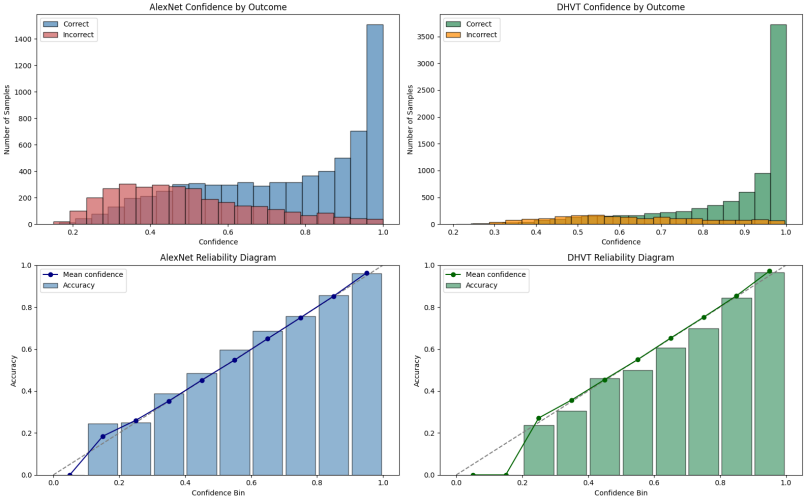


True: deer
AlexNet: airplane
DHVT: airplane



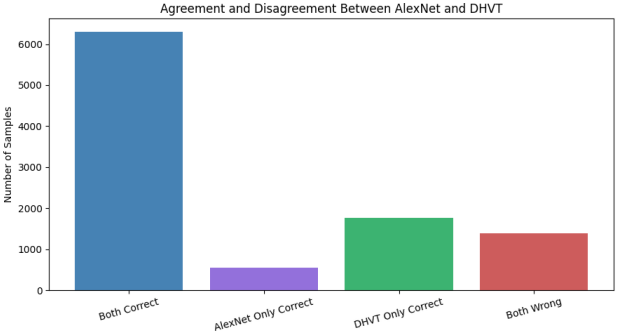
True: airplane
AlexNet: bird
DHVT: frog

Confidence Distribution



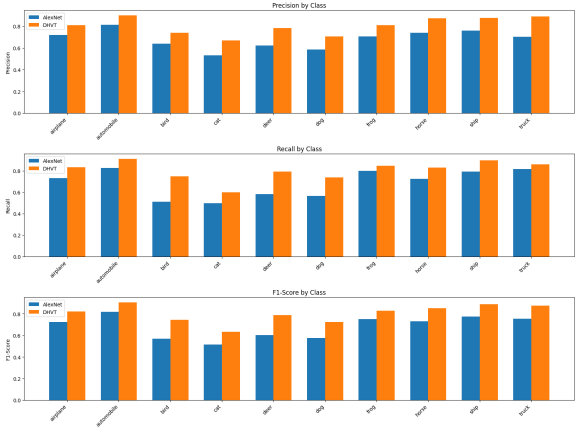
- DHVT is more confident overall, but the figure also shows how confidence behaves on

Agreement Analysis



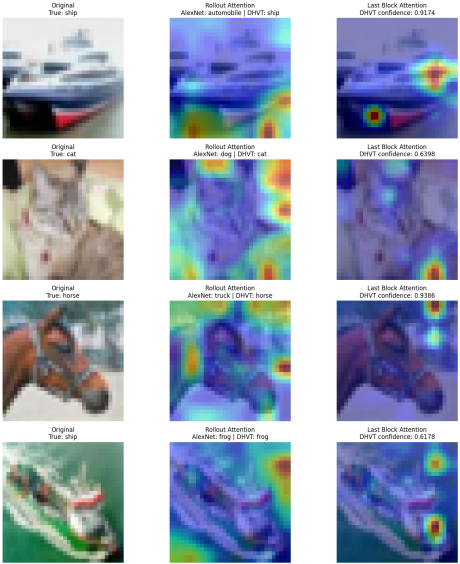
Case	Ratio
Both correct	0.6304
AlexNet only	0.0544
DHVT only	0.1758
Both wrong	0.1394

Precision, Recall, and F1



- DHVT stays ahead on precision, recall, and F1, so its advantage is not metric-specific.
- This confirms stronger class discrimination and better balance between false positives and false negatives.

DHVT Attention Visualization



Conclusion

- DHVT is the stronger model in this experiment.
- It improves validation accuracy, macro metrics, and calibration while using fewer parameters.
- The gain is visible in curves, class-level results, agreement analysis, and attention visualization.
- A fair comparison needs multiple views of evaluation, not only one final accuracy number.

Future Work

- Extend the comparison to another dataset such as CIFAR-100 or a custom image folder.
- Add a dedicated calibration slide with reliability-diagram comparison for both models.
- Study whether the same pattern remains true under longer training or data augmentation changes.

References

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